

Boo Wyane

Applied AI Engineering | Python & LLM Agents

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PROFILE

Astrophysics graduate building Python-based AI agents and computer-vision applications. Hands-on experience with structured LLM outputs, retrieval and persistent memory, alongside data-migration UAT and operations analytics. Seeking junior AI engineering roles.

AI AND SOFTWARE PROJECTS

Open Agent Hackathon 2026 | Reasoning AI Agent

- Developed Python pipelines integrating Malaysian weather forecasts and PM2.5/PM10 air-quality data into Zetaris, exposing district-level SQL views for haze analysis.
- Designed a four-agent architecture for outdoor-work planning, incorporating versioned evidence, persistent memory, and independent critic review to validate recommendations.

Staff Badge Detection and Tracking

- Implemented a modular Python video pipeline using OpenCV, Roboflow and ByteTrack to detect badges and track people across frames.
- Associated badge detections with person bounding boxes using overlap and confidence thresholds; exported annotated video and frame-level CSV/JSON results.

Particle Detector Simulation | C++

- Designed a C++ particle detector simulation using abstract classes and polymorphism to model particles, trackers, calorimeters and muon detectors.
- Implemented input validation for non-negative particle energies and checked that assigned energies matched each event's total input energy.

Scientific Data Analysis

- Analysed Higgs boson decay in CERN ATLAS data using ROOT; studied stellar signals through astronomical data analysis at Jodrell Bank Observatory.

TECHNICAL SKILLS

AI engineering: Structured LLM outputs, Pydantic, planning/evaluation, persistent memory, embeddings, hybrid retrieval, Multi-agent architecture, evidence validation, agent memory design

Computer vision: OpenCV, Roboflow, ByteTrack

Programming: Python, SQL, C++, JavaScript, Java, MATLAB, R

Data: Pandas, NumPy, SQLite, data validation and visualisation

Data engineering: Data ingestion pipelines, REST API integration, environmental data processing

Data platforms: Zetaris

Infrastructure: Docker, Docker Compose, PostgreSQL configuration

Testing and reliability: pytest, UAT, transactions, idempotency, provenance, rollback

Development: Git, GitHub, React Native (Expo), HTML, CSS, Expo CLI, EAS Build

Analytics tools: BigQuery, Looker Studio, Tableau, Salesforce, Excel, Google Sheets, Notion

PROFESSIONAL EXPERIENCE

Great Eastern — Agency Operations Analyst (Jul 2025 – Jun 2026)

- Implemented digital process improvements using Looker and Notion to support agency operations and operational efficiency.
- Designed a strategic improvement plan aligned with an annual ANP sales target of RM2M.

Philip Morris International — Data Migration & Automation (UAT) (Feb 2025 – Mar 2025)

- Identified name-parsing edge cases caused by Malaysian naming structures during data migration and user acceptance testing.
- Coordinated with technical and business teams on migration testing and deployment readiness.

Pearson — Student Ambassador (Jun 2022 – Jun 2023)

- Conducted student market analysis and engagement activities, collaborating with marketing teams on student outreach.

Private Academic Tutor (Aug 2022 – Jul 2023)

- Designed tailored physics learning plans that helped students improve from failing to passing grades.

EDUCATION AND CERTIFICATIONS

IBM AI Engineering Professional Certificate | Coursera (In progress)

Completed coursework: Machine Learning with Python; Introduction to Deep Learning & Neural Networks with Keras.

Google AI Professional Certificate | Coursera (Aug 2026)

Google Data Analytics Professional Certificate | Coursera (Jan 2026)

University of Manchester — BSc Astrophysics (2021 – 2024)

Upper Second-Class Honours (2:1).

LEADERSHIP AND ACTIVITIES

Student Mentor (Sep 2022 – Jun 2023)

Led weekly sessions for first-year students on study methods and academic strategies.

President — Chinese Orchestra (Jun 2018 – Jun 2019)

Led a 100-member orchestra to a national gold medal; coordinated rehearsals and performances.

LANGUAGES

English: Fluent

Mandarin: Native

Malay: Intermediate

Cantonese: Fluent